Data + Questions

Observational unit: one row/record. **Population**: broader units/process of interest; observed data are not automatically the population. **Group**: category used to compare. **Outcome**: measured response. **Context**: structure to preserve (e.g., region/weekend). **Point estimate**: one statistic from observed data.

Workflow: question + unit → group/outcome/context → inspect shape, types, missingness, labels → choose statistic before testing → summarize/visualize → state sampling/dependence limits → simulate reproducibly → interpret in original units.

Description ≠ causation: an observed group difference does not prove group membership caused it. Generalization needs representative sampling; causation needs appropriate design/assumptions.

Statistic	Meaning
Mean diff.	$\bar{x}_A - \bar{x}_B$: average magnitude; sensitive to extremes.
Median diff.	$\tilde{x}_A - \tilde{x}_B$: middle; less outlier-sensitive.
Rate diff.	$\hat{p}_A - \hat{p}_B$: fraction above threshold/Boolean condition.
Quartile diff.	$Q_{.75,A} - Q_{.75,B}$: upper distribution, not maximum.

Always state subtraction order, units, and missing-value handling. Mean of a Boolean condition = proportion true: `(x > c).mean()`.

Permutation Tests

Purpose: test evidence against a no-association null model. Under H_0, outcomes and labels are unrelated; reassignment creates null worlds.

1. Choose question-matched statistic T.
2. Calculate observed T_{obs}.
3. Identify what must remain fixed (group sizes; strata if relevant).
4. Reassign labels under H_0; recompute T.
5. Null distribution = simulated statistics; compare to T_{obs}.

Exchangeability: units may be shuffled only when the null makes them interchangeable. Stratify/permutate within region or other design cells when that structure must remain fixed. Ask: what relationship is broken, and what is preserved?

$$\hat{p} = \frac{\#\{T^* \text{ at least as extreme as } T_{obs}\} + 1}{B + 1}$$

Two-sided: extreme in either direction, usually $|T^*| \geq |T_{obs}|$. Resolution floor $\approx 1/(B+1)$. A p-value is not $P(H_0 \mid \text{data})$ and not the probability the result happened "by chance."

Interpretation: Under the null model represented by the procedure, about p of reassignments produced a statistic at least as extreme as observed; this is [weak/strong] evidence against no association for these units, subject to exchangeability and sampling assumptions.

Bootstrap

Purpose: estimate sampling variability by treating observed data as an empirical population. Resample observed units **with replacement**, usually separately within groups; preserve each group sample size.

Point estimate: original statistic. **Bootstrap sample**: one resampled dataset. **Replicate**: statistic from one bootstrap sample. **Distribution**: all replicates. **SE**: SD of replicates. **95% percentile CI**: $[q_{.025}, q_{.975}]$.

Bootstrap Interval: If the recorded groups reasonably represent their target populations and the resampling assumptions are reasonable, the interval gives plausible values for the population difference under the method's long-run coverage interpretation. It is not the probability that a fixed parameter is random, and it does not repair biased or unrepresentative data.

Permutation	Bootstrap
Test no association	Estimate variability/uncertainty
Null world; labels reassigned	Empirical world; values resampled
Typically no value replacement	With replacement
p-value	SE and CI
Exchangeability under H_0	Data represent target population

Structured bootstrap: resample within design cells (e.g., visitor type × weekend) when composition/structure should be preserved. Compare ordinary vs structured as a sensitivity check. A CI does not repair bias or mean a fixed parameter has a probability of lying in this interval.

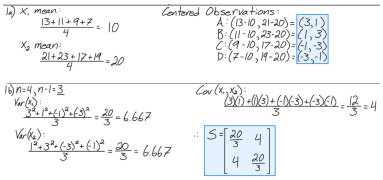
Distance + Scaling

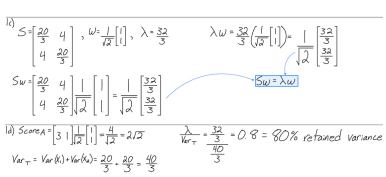
Feature representation, units, scale, and order determine distance. Remove an anchor's self-distance before ranking; self-distance is 0. Standardize, then recompute distances.

$$d_E(x,y) = \sqrt{\sum_j (x_j - y_j)^2}, \qquad d_M(x,y) = \sum_j |x_j - y_j|$$

$$z_j = \frac{x_j - \bar{x}_j}{s_j}; \qquad \text{HW2A: } s = \texttt{std(ddof=0)}$$

Large-scale features can dominate raw Euclidean distance; standardization makes feature scales comparable but may remove meaningful domain scale. Manhattan is less outlier-amplifying than Euclidean.





Similarity

$$s_{cos}(x,y) = \frac{x^T y}{\|x\|_2 \|y\|_2}, \qquad \|x\|_2 = \sqrt{\sum_j x_j^2}$$

Cosine compares direction/proportions and ignores overall magnitude; zero vectors have undefined direction. If a library returns cosine distance, $s = 1 - d$.

$$J(A,B) = \frac{|A \cap B|}{|A \cup B|}$$

Jaccard uses binary presence sets: ignores quantities and shared absences; 1 identical, 0 no shared presence. No metric is universally best; match metric to the meaning of similarity.

PCA Foundations

Dimensionality reduction: represent observations with fewer coordinates. **Selection** keeps original features; **extraction** creates new features. PCA is linear feature extraction for correlated/redundant measurements.

$$s^2 = \frac{1}{n-1}\sum_i (x_i - \bar{x})^2, \qquad \text{Cov}(x,y) = \frac{1}{n-1}\sum_i (x_i - \bar{x})(y_i - \bar{y})$$

$$S = \begin{bmatrix} \text{Var}(x_1) & \text{Cov}(x_1,x_2) \\ \text{Cov}(x_1,x_2) & \text{Var}(x_2) \end{bmatrix}$$

Diagonal = variances; off-diagonal = covariances. Total variance = $\text{tr}(S) = \sum_j \lambda_j$.

Eigen equation:

$$Sw = \lambda w, \qquad \|w\|_2 = 1, \qquad z = w^T x_c$$

w = eigenvector/loadings (direction); λ = variance captured. PC1 maximizes projected variance; later PCs are orthogonal and maximize remaining variance. Component order: descending λ. Flipping all signs of a component changes no meaning.

$$\text{EV fraction}_k = \frac{\lambda_k}{\sum_j \lambda_j}, \qquad \text{cumulative retention}_K = \frac{\sum_{k=1}^{K} \lambda_k}{\sum_j \lambda_j}$$

Choose K by required retention and cost of losing detail; a 90% rule is not universal. Standardize first when units/scales differ; HW2B uses sample s (`ddof=1`).

Term	Meaning
Loading	Coefficient/direction; `pca.components_`.
Score	One record's component coordinates; `pca.transform(X)`.
Eigenvalue	Variance along a component; `explained_variance_`.
EV ratio	Variance fraction; `explained_variance_ratio_`.
Reconstruction	Reduced scores → standardized feature space via `inverse_transform`; undo scaling for original units.

Loading interpretation: compare sign/magnitude within one component. A score's sign indicates alignment with the chosen direction; eigenvector sign is arbitrary. High total retention need not mean every feature/individual reconstructs equally well. New data: use saved training means/scales, then `transform`; never refit on the new row.

Code + Concept Traps

- `True` behaves as 1, `False` as 0 in a mean; check missing values first.
- Permutation shuffles labels/assignments; bootstrap resamples values/units with `replace=True`.
- Seed once before the simulation loop; reseeding inside repeats the same first draw.
- Independent groups are not paired observations; choose the appropriate permutation type.
- `cdist(..., metric="cosine")` returns distance, not similarity; use $1 - d$.
- Double brackets such as `matrix.loc[[anchor]]` preserve a 2D row for `cdist`.
- Remove `anchor` before sorting nearest neighbors.
- Raw numeric, standardized numeric, and binary presence/absence are different representations.
- `fit` learns directions; `transform` projects; `components_` are loadings, not scores.
- Check `ddof` = 0 vs `ddof` = 1; it changes scales, covariance, and PCA calculations.

Fast Interpretation Starters

- **Difference**: "$A - B$ is positive/negative, so A is larger/smaller in [units]."
- **p-value**: "Under [specific null/simulation], p of simulated statistics were at least this extreme."
- **CI**: "The method gives plausible population values from [lower] to [upper], assuming [sampling/resampling assumptions]."
- **Distance**: "Smaller means closer under this representation and metric; it does not imply universal similarity."
- **Similarity**: "Larger means more similar according to [direction / shared presence], not necessarily larger magnitude."
- **PCA**: "This loading shows the feature's direction/relative contribution; this score locates one observation in component space."

Pre-Answer Checklist

What is one row? What are group/outcome/context? What statistic and subtraction order? What is randomized, and what is fixed? With or without replacement? Raw, standardized, or binary? Distance or similarity? Anchor removed? Which s convention? Does output describe one row, all rows, or a component? What assumptions limit interpretation?